AOS-CX 10.18.xxxx Containers Guide (6300F/M, 6400, 8xxx, 9xxx, 100xx Switch Series)

Published: May 2026

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Container overview

The Container Manager feature allows containers to run inside an AOS-CX switch. It allows the switch to launch multiple container images, whether HPE-signed or unsigned, and provides tools to manage, maintain, and monitor the lifecycle of these images and containers. Applications can be run within the container infrastructure, turning them into containerized applications.

The AOS-CX container infrastructure is designed to support containers that run in the background as services or scheduled jobs. These containers either keep running continuously or stop gracefully after completing their tasks. Examples of supported workloads include web servers like Nginx or Apache, application servers built with Node.js, Java, or Python APIs, and system monitoring tools.

The container infrastructure does not support containers that must interact users in real time through a command-line interface, such as shells (bash, sh), interpreters (Python, Node.js), or text-based tools (like vim or top) to accept user input and display output interactively. If you try to deploy an unsupported container that relies on an interactive terminal, it may stop immediately or behave unpredictably.

Subtopics

Supported switches

The following switch platforms are validated for data center deployments. The agent itself has no context about the switch that it is installed on, except that the chipset architecture must be either **x86_64** or **arm**. The container image size is limited by platform. A container instance will display an error if you attempt to download a container image that exceeds the supported image size.

Switch	Chipset Architecture Type	Maximum Image size (MiB)
6300	arm	500
6400	arm	500
8100	arm	500
8320	x86_64	1000
8325	x86_64	1000

Switch	Chipset Architecture Type	Maximum Image size (MiB)
8325H	x86_64	1000
8325P	x86_64	1000
8360	arm	500
8400	x86_64	500
9300	x86_64	1000
9300S	x86_64	1000
10000	x86_64	1000

NOTE

The HPE Aruba Networking CX vSphere agent is a containerized agent developed for data center applications. The plugin consumes memory and CPU that would otherwise be allocated to the switch. Thus, the targeted platforms have higher system resources. Those resources are capped to prevent its interference with performance of the switch.

NOTE

The solution was tested and qualified on the platforms listed here. Other platforms may work at low scale, but they are not recommended outside of the lab or test environments due to the resource requirements for vSphere.

USB persistent storage

About this task

Containers can mount additional persistent storage via a USB device using the mount, source and destination commands to create a mount ID and specify the mount source and destination paths. Best practices is to configure persistent storage via these commands. If a USB directory in the source filesystem is deleted or renamed through shell commands, the container may not function correctly.

The USB persistent storage feature includes the following additional caveats.

- Containers are not aware if the configured USB device is unmounted, disabled or physically removed. As a result, the container's correct functionality can't be guaranteed if these actions are made without unconfiguring the container mount first.

- For containers using USB persistent storage functionality, data won't synchronize between the management modules after a High Availability event, so the container deployed on the new management module won't have access to the USB device connected to the previous management module.
- If there is an issue with the USB device mount when the container boots up, there can be a potential wait of up to two minutes before the container shows an error.

The following steps configure a container with USB persistant storage.

Procedure

1. Create the container context

```
switch(config)# container <NAME>
```

2. Configure the container image location

```
switch(config-container-<NAME>)# image-location <URL> vrf <VRF>
```

3. Configure the container mount

```
switch(config-container-<NAME>)# mount <ID>
```

4. Configure the container mount source

```
switch(config-container-mount-<ID>)# source usb /app1-data
```

5. Configure the container mount destination

```
switch(config-container-mount-<ID>)# destination /app-data
```

6. Enable the container

```
switch(config-container-<NAME>)# enable
```

7. Validate the container status

```
show container <NAME>
```

Restart protection

The HPE Aruba Networking switch can prevent a crashing containerized application from restarting after three restarts within a brief period of time. Once the restart protection feature indicates that it has stopped a crashing application in this manner, the container will display a **run failed** status in the output of the **show container** command.

If the configured image has been downloaded and verified by container manager, in case of a restart, the image will not need to be downloaded again unless the image location is modified. This also applies when using the **allow-unsigned** parameter, if a re-download is required, modify the image location URL or remove the command and add it again.

Network modes

Containers can be configured to run in one of three supported network modes:

- **Default mode**. In the default mode there is no network connectivity. This option is used when there is no network configuration specified, and does not allow the container network connectivity to reach external devices.
- **VRF network mode**. This mode allows network access through one of the switch VRFs. Communications between container and the host switch are restricted to the default management interfaces ports (22 and 443).
- **Isolated network mode**. The 8325, 8325H and 8325P Switch series support isolated network mode, which allows network access through one or more configured VLANs. Containers using this mode behave as hosts physically connected to the switch's front panel ports with a unique MAC address assigned by the system. This allows the container configuration to support VLAN-aware configurations and Ethernet traffic types (unicast, broadcast, multicast). Note that this mode does not allow communication with the switch interfaces and services. For more information on configuring a container in isolated network mode, see Configuring an isolated network.

Configuring an isolated network

Configuring an isolated network on a 8325, 8325H or 8325P Switch series is a two-part process. You must create the isolated network first, then configure the container with isolated network connectivity.

Configuring the isolated network

1. If it is not already created, create the VLAN to be used by the isolated network.
```
switch(config)# vlan 789
```
2. Configure the isolated network.
```
switch(config)# container isolated-network test
```
3. Create the virtual interface
```
switch(config-container-isolated-network-test)# virtual-interface vlan
test
```

4. Set the IP address for the interface.

```
switch(config-container-isolated-network-virtual-interface)# ip address
90.0.0.1/30
```

5. Enter **exit** to return to the **config-container-isolated-network-<NAME>** context, then configure the default gateway.

```
switch(config-container-isolated-network-virtual-interface)# exit
switch(config-container-isolated-network-test)# default-gateway 90.0.0.2
```

6. Enable mDNS.

```
switch(config-container-isolated-network-test)# mdns enable
```

7. Exit the **config-container-isolated-network** context and validate the status of the isolated network.

   The following example verifies an isolated network with virtual interface that uses VLAN 789,

```
switch(config-container-isolated-network-<NAME>)# exit
switch (config)# vlan 789
switch (config-vlan-789)# exit
switch(config)# container isolated-network test
switch (config-container-isolated-network-test)# virtual-interface vlan
789
switch (config-container-isolated-network-virtual-interface)# ip address
90.0.0.1/30
switch (config-container-isolated-network-virtual-interface)# exit
switch (config-container-isolated-network-test)# default-gateway 90.0.0.2
switch (config-container-isolated-network-test)# mdns enable
switch (config-container-isolated-network-test)# end
switch # show container isolated-network test
Isolated network        : test
Status                  : network_provisioned
Status reason           : n/a
Default gateway         : 90.0.0.2
mDNS                    : enabled
Virtual MAC             : 64:e8:81:1a:4f:07
Containers              : n/a
Virtual interfaces      :
------------------------
VLAN  IP address
------------------------
789   90.0.0.1/30
```

Connect the container to the isolated network

1. Create the container context.

```
switch(config)# container test
The feature being used requires a AOS-CX Advanced Software Feature Pack.
For more information, refer to the AOS-CX Feature Pack Deployment Guide.
```

2. Configure the container network mode.

```
switch(config-container-test)# network isolated-network test
```

3. Configure the container image location.

```
switch(config-container-test)# image-location http://117.0.0.2:8000/
vsphere_agent-1.3.0-96-x86_64.img vrf mgmt
```

4. Enable the container, then exit the container context

```
switch(config-container-test)# enable
switch(config-container-test)# end
```

5. Validate the container status.

```
switch # show container test
Container                : test
Container status         : operational
Manifest status          : success
Image status             : verified
Image version            : 1.3.0-96
Image location VRF       : mgmt
Image location URL       : http://117.0.0.2:8000/vsphere_agent-1.3.0-96-
x86_64.img
CPU limit                : 10%
Memory limit             : 256MB
VRFs                     :
Environment variables    :
Encrypted environment variables:
VRF networks: n/a
Isolated network: test
Mounts: n/a
```

Example configuration:

```
(switch)(config)# vlan 789
(switch)(config-vlan-789)# exit
(switch)(config)# container isolated-network test
(switch)(config-container-isolated-network-test)# virtual-interface vlan 789
(switch)(config-container-isolated-network-virtual-interface)# ip address
90.0.0.1/30
(switch)(config-container-isolated-network-virtual-interface)# exit
(switch)(config-container-isolated-network-test)# default-gateway 90.0.0.2
(switch)(config-container-isolated-network-test)# mdns enable
(switch)(config-container-isolated-network-test)# end
```

```
(switch)# show container isolated-network test
Isolated network           : test
Status                    : network_provisioned
Status reason             : n/a
Default gateway           : 90.0.0.2
mDNS                      : enabled
Virtual MAC               : 64:e8:81:1a:4f:07
Containers                : n/a
Virtual interfaces        :
------------------------
VLAN  IP address
------------------------
789   90.0.0.1/30
```

vSphere agent

About this task

The HPE Aruba Networking CX vSphere agent allows administrators to trace how Virtual Machines are connected to the physical switches. The ability to visualize how a Virtual Machine connects to a CX port helps to resolve connectivity issues. This functionality helps operators in several troubleshooting scenarios, such as diagnosing a connectivity issue, or ensuring adequate distribution of VMs across the physical infrastructure.

You can install the vSphere agent on one or many of the switches in the fabric. You can perform the following tasks, from any switch where the agent is installed:

Procedure

1. Run the show VMs command to list all the VMs managed by the vCenter Appliance integrated with the agent. This functionality also helps to view VMs connections to the ports on other switches on the fabric.
2. Search for a specific VM by name. To find the connection between the VM and a switch port, attach the switch to at least one host managed by the integrated vCenter.

Results

> **NOTE**
>
> The **show neighbors** command examines the LLDP data that is advertised from any switch to a distributed virtual switch and provides connection details for third-party switches.

For technical information on containers, including use cases, example configurations and best practices, view the **AOS-CX Container Enhancements video on the HPE Aruba Networking broadcasting channel on YouTube**. For more information about cloud capabilities of vSphere agent, refer to .

Subtopics

Installation and Configuration of the vSphere agent

About this task

To install a container, perform the following steps:

Procedure

1. Download the vSphere agent from the HPE Aruba Networking support portal and select the **arm** or **x86_64** architecture for the switch platform. Refer to the list of supported switches to select the corresponding architecture.
2. Host the agent on a local network that can be accessed using the HTTP protocol. Limitations of the container framework prevent usage of the HTTPs protocol. The HTTP server must be reachable on the mgmt or any other configured VRF.

Results

To create the container configuration:

1. Enter the **config** command to enter the config context.
2. Create and enter the **config-container-<name>** context using the container command.
3. From the **config-container-<name>** context, configure the container image location using the image-location command. To create a container configuration that bypasses signature validation, include the optional **allow-unsigned** parameter for the image-location command
4. (Optional) Configure a container environment variable using the env command.
5. (Optional) To create a configuration with VRF network connectivity, issue the network vrf command, then configure port mapping using the port-map command.
6. (Optional) To create a configuration that limits CPU usage to a percentage of the total switch CPU, use the restrict cpu command.
7. (Optional) To create a configuration that limits memory resources to maximum number of megabytes (up to 20% of the total switch memory), use the restrict memory command.

8. (Optional) To create a configuration with mounted storage, issue the mount command, then use the source and destination parameters from within the **config-container-mount-<id>** context to define the mount source and destination.
9. From the **config-container-<name>** context, enable the container using the enable command.
10. Issue the show container command to validate the container status.

vSphere show commands

There are three top-level show commands.

- vms - shows virtual machine connection information.
- connection-status - shows information about configured vSphere vCenters.
- neighbors - shows learned neighbor information (Note: This information is learned from LLDP on vCenter distributed vswitches, not the switch).

show vms

The following show command describes the status of the connection to vSphere, and show the VMs that are attached to switches:

At least one command parameter must be used if the parameters are available.

For example, with the **show vms** command, we force use of the **brief** option so the user is aware of all available options.

If not specified, all commands default to **brief**.

```
igor-sw-02# container vsphere exec show vms brief Hypervisor VM Name
Physical Switch Connections Interface
laser-02-esxi.lab.plexxi.com vm4 laser-sw-01 (b8:d4:e7:d9:af:00) 1/1/3 vm2
laser-sw-02 (b8:d4:e7:da:40:00) 1/1/4
laser-sw-02 (b8:d4:e7:da:40:00) 1/1/3 vm5 laser-sw-02 (b8:d4:e7:da:40:00)
1/1/4 laser-sw-02 (b8:d4:e7:da:40:00) 1/1/3 laser-sw-01 (b8:d4:e7:d9:af:00)
1/1/3
laser-01-esxi.lab.plexxi.com vm1 laser-sw-01 (b8:d4:e7:d9:af:00) 1/1/1 vm3
laser-sw-01 (b8:d4:e7:d9:af:00) 1/1/2
laser-sw-02 (b8:d4:e7:da:40:00) 1/1/1
igor-sw-02# container vsphere exec show vms vm-name vm4 Hypervisor VM Name
Physical Switch Connections Interface
laser-02-esxi.lab.plexxi.com vm4 laser-sw-01 (b8:d4:e7:d9:af:00) 1/1/3 igor-
sw-02# container vsphere exec show vms vm-name vm4 detailed VM Name : vm4
```

```
Hypervisor : laser-02-esxi.lab.plexxi.com Vendor : vmware
Hostname : vm4
OS : CentOS 4/5 or later (64-bit) VM status : on
Virtual NIC : Network adapter 2 IP Address : 172.20.12.204
MAC Address : 00:50:56:89:3c:bf Virtual Switch : dvs4
Port Group : dpg4 Vlan : 0
Physical Switch Connections Interface : 1/1/3
Chassis ID : b8:d4:e7:d9:af:00 Switch Name : laser-sw-01
Switch Description : HPE_ANW JL635A GL.10.12.0001G Management Address :
172.20.11.253
Container exec commands are top-level, and can be run in any CLI context.
igor-sw-02# container vsphere exec show connection-status brief Hypervisor
Last Sync Connection State
laser-vcsa.lab.plexxi.com 2023-03-17 23:30:03 connected igor-sw-02# config
igor-sw-02(config)# container vsphere exec show connection-status brief
Hypervisor Last Sync Connection State
laser-vcsa.lab.plexxi.com 2023-03-17 23:30:07 connected igor-sw-02(config)#
interface 1/1/1
igor-sw-02(config-if)# container vsphere exec show connection-status brief
Hypervisor Last Sync Connection State
laser-vcsa.lab.plexxi.com 2023-03-17 23:30:15 connected
```

show connection-status

This command displays the connectivity status for the vSphere agent.

The connection timestamp is updated whenever a successful connection is made with the vSphere agent. The sync timestamp is updated whenever a full sync, event poll, or event process occurs.

```
igor-sw-02(config-container-vsphere)# container vsphere exec show
connection-status brief Hypervisor Last Sync Connection State
laser-vcsa.lab.plexxi.com 2023-03-18 02:36:06 connected
igor-sw-02(config-container-vsphere)# container vsphere exec show
connection-status detailed Hypervisor : laser-vcsa.lab.plexxi.com
Connection State : connected Last Sync : 2023-03-18 02:35:08
Last Connection : 2023-03-18 02:35:09
igor-sw-02(config-container-vsphere)# container vsphere exec show
connection-status detailed Hypervisor : laser-vcsa.lab.plexxi.com
Connection State : disconnected Last Sync : 2023-03-18 02:04:05
Last Connection : 2023-03-18 02:09:30
Connection Failure Reason : Could Not Resolve Hostname
```

show neighbors

show neighbors command displays the virtual network adapter neighbor information.

Neighbors are pulled from distributed virtual switches of the vSphere and cached locally. Neighbors are used to map VMs to physical switch ports.

If you do not see a neighbor on the port, you will not see any VM connectivity on that port either. Detailed and brief options are available. Filtering can be done on switch-name and/or interface.

```
igor-sw-02(config-container-vsphere)# container vsphere exec show neighbors
brief Chassis ID Switch Name Interface
b8:d4:e7:da:40:00 laser-sw-02 1/1/1 1/1/2
b8:d4:e7:d9:af:00 laser-sw-01 1/1/1 1/1/2
igor-sw-02(config-container-vsphere)# container vsphere exec show neighbors
detailed Chassis ID : b8:d4:e7:da:40:00
Switch Name : laser-sw-02
Switch Description : HPE_ANW JL635A GL.10.10.1000 Management Address :
192.168.200.1
Interface : 1/1/1 Advertisement Type : lldp TTL : 102
Interface : 1/1/2 Advertisement Type : lldp TTL : 102
Chassis ID : b8:d4:e7:d9:af:00 Switch Name : laser-sw-01
Switch Description : HPE_ANW JL635A GL.10.10.1000 Management Address :
192.168.200.0
Interface : 1/1/1 Advertisement Type : lldp TTL : 102
Interface : 1/1/2 Advertisement Type : lldp TTL : 102
```

Container management commands

Select a command from the list in the left navigation menu.

Subtopics

- **container**
- **container exec**
- **default-gateway**
- **enable**
- **env**
- **image-location**
- **ip address**
- **mdns enable**
- **mount**
- **network isolated-network**
- **network vrf**
- **port-map**
- **restrict cpu**
- **restrict memory**
- **show container**
- **show capacities containers**
- **show capacities-status containers**
- **show running-config container**

virtual-interface vlan
vrf

`container`

Syntax

```
container <CONTAINER-NAME>|{isolated-network <name>}
no container <CONTAINER-NAME>
```

Description

Enters into the container configuration context. On 8325, 8325H and 8325P Switch series, include the optional isolated-network parameter to configure containers using this network to appear as hosts physically connected to the switch's front panel ports.

The **no** form of this command removes the existing configurations of the specified container.

Parameter	Description
`<CONTAINER-NAME>`	**The container name can contain up to 64 characters. Only letters, numbers and underscore (_) characters are permitted.**
`isolated-network <name>`	(For 8325, 8325H and 8325P Switch series) Use this parameter to configure a container isolated network supporting VLAN - aware configurations and unicast, broadcast, and multicast Ethernet traffic types. The isolated network name can contain up to 32 characters. Only letters, numbers and underscore (_) characters are permitted.

Usage

Note the following considerations when configuring an isolated network.

- You can create only one isolated network configuration, and this configuration will apply to all containers operating in isolated network mode.
- Multicast traffic only works with the multicast DNS (mDNS) protocol.
- Containers can use native ports, so you do not need to configure port mapping.
- Be careful to avoid port conflicts between containers when running containers in this mode.
- If you change the settings of an isolated network used by a container, the container will restart to apply those changes.

Example

Configures a new container:

```
switch(config)# container app
The feature being used requires a AOS-CX Advanced Software Feature Pack.
For more information,refer to the AOS-CX Feature Pack Deployment Guide.
```

AOS-CX does not enforce the requirement to own a feature pack prior to using container features. This warning message is displayed only during creation, subsequent calls to the container context will not display the message.

Setting the container configuration to isolated network mode:

```
switch(config)# container isolated-network net1
```

Command History

Release	Modification
10.12	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8325H 8325P 8360 8400 9300 9300S 10000 10040	`config` `config-container-<CONTAINER-NAME>`	Administrators or local user group members with execution rights for this command.

`container exec`

Syntax

```
container <NAME> exec <PARAMS>
```

Description

Allows the execution of an endpoint script in the container. The location of this endpoint is provided to the container manager infrastructure through a manifest file in the image file system of the container. This manifest file provides metadata related to the container application. When the exec command runs, the manifest information is used to determine the endpoint to execute and the user parameters are passed directly to the endpoint. The output of such execution is provided directly to the user through the CLI. In case the manifest information or the endpoint file are missing an error is presented to the user. The user can interrupt the execution by pressing **Ctrl+C**.

> **NOTE**
> If the container is not operational when the command is executed, the following error message is returned: **Failed to execute endpoint - The container is not operational.**

Parameter	Description
`<NAME>`	Specifies a container name up to 64 characters long.
`exec`	Runs a container application command.
`<PARAMS>`	Specifies container command parameters.

Command History

Release	Modification
10.12	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8325H 8325P 8360 8400 9300 9300S 10000 10040	`config-container-<CONTAINER-NAME>`	Administrators or local user group members with execution rights for this command.

`default-gateway`

Syntax

```
default-gateway <ip-addr>
no default-gateway [<ip-addr>]
```

Description

Configures a default gateway IP address for the isolated network. The default gateway is used by containers to route traffic outside the local network provided by the virtual interfaces. After completing the configuration, use the `show container isolated-network` command to verify that the default gateway is valid. It will automatically check the IP address is within the subnet of at least one configured virtual interface and report any error.

The **no** version of this command removes the default gateway configuration.

Parameter	Description
`<ip-addr>`	The default gateway used by a containers in isolated network m ode to route traffic outside the local network provided by the vi rtual interfaces.

Example

Configuring a default gateway:

```
switch(config-container-isolated-network-<name>)# default-gateway <ip-addr>
```

Command History

Release	Modification
10.17	Command introduced

Command Information

Platforms	Command context	Authority
8325 8325H 8325P	`config-container-isolated-network-<network-name>`	Administrators or local user group members with execution righ ts for this command.

`enable`

Syntax

```
enable
no enable
```

Description

Enables the container. By default, a container is disabled when it is created, meaning it cannot execute. Enabling the container allows it to start execution. Once a container is enabled, any subsequent configuration change causes it to restart so the new configuration can be applied. For this reason, it is considered best practice to complete as much configuration as possible before enabling the container.

If a container is enabled and the switch it is running on is restarted, the container will automatically start again after the switch reboots, without requiring any additional user intervention.

The **no** version of this command disables the container. When an enabled container is disabled, system resources are released but the container configuration settings are stored.

Example

Enabling and then disabling the container **test**:

```
switch(config-container-test)# enable
switch(config-container-test)# no enable
```

Command History

Release	Modification
10.17.1000	Additional feature information.
10.16.1000	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8325H 8325P 8360 8400 9300 9300S 10000 10040	`config-container-<CONTAINER-NAME>`	Administrators or local user group members with execution rights for this command.

env

Syntax

```
env <NAME> {value <VALUE>}|{encrypted [plaintext <VALUE>|ciphertext
<VALUE>]}
no env <NAME> {value <VALUE>}|{encrypted {plaintext|ciphertext}<VALUE>}
```

Description

Configures an environment variable for a container that is composed of a key and a value pair. The key-value pair defines the behavior of the environment in a container and is used by the container processes. The value of the environment variable can be stored in the host system as an encrypted value. The container manager infrastructure provides the decrypted value to the container.

The **no** form of this command removes the configured environment variable from a container.

> **NOTE**
>
> The predefined environmental variable **CX-SERIAL** is used to share for the switch serial number to a container and is reserved for internal use only. Do not modify this variable, or attempt to create another variable with the same name.
>
> Configuring the **env** variable for an already operational container causes the container to restart if the container has already been enabled.

Parameter	Description
`<NAME>`	Specifies the name of the container environment variables.
`value <VALUE>`	Specifies the variable value.
`encrypted`	Encrypts the environment variable value. If you press **<enter>** a fter the **encrypted** parameter, you will enter a variable configur ation mode that allows you to securely enter a hidden value. Thi s is the recommended method for entering an encrypted variabl e

Parameter	Description
`plaintext <VALUE>`	**Optional**. Specifies the variable value in plain text. Not recomm ended for encrypted variables.
`ciphertext <VALUE>`	**Optional**. Specifies the variable value as previously encrypted text. Recommended for encrypted variables; specify the encry pted variable value as previously encrypted text.

Example

Securely entering an encrypted variable:

```
6300(config-container-test)# env TEST encrypted
Enter environment variable value: ********
6300(config-container-test)# end
```

Command History

Release	Modification
10.13.1000	The **plaintext** and **ciphertext** options for the encrypted parameter are now opti onal. Starting with this release, you can use the **encrypted** option to encrypt the environment variable and specify the value in plaintext hidden from the CLI.
10.12	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8325H 8325P 8360	`config-container-<CONTAINER-NAME>`	Administrators or local user group members with execution righ ts for this command.

Platforms	Command context	Authority
8400		
9300		
9300S		
10000		
10040		

`image-location`

Syntax

```
image-location <URL> vrf <VRF-NAME> [allow-unsigned]
no image-location <URL>
            <VRF-NAME> [allow-unsigned]
```

Description

Configures the image location for a container. Modifying image location prompts an image upgrade.

The **no** form of this command removes the configured location of a container.

> **NOTE**
>
> - If the user sets a location value which does not follow the standard URL format, the following error message is returned: **Failure to configure image location: Invalid URL**
> - If the user tries to use a VRF value that doesn't exist on the switch, the following error message is returned: **Failure to configure image location: Invalid VRF**
> - If the image of the container exceeds the maximum image size supported by the switch, the container won't be deployed.

By default, only container images with a valid HPE signature are allowed. To bypass this signature check and allow unsigned container images, include the **allow-unsigned** parameter when you define the image location. The **allow-unsigned** parameter cannot be used if you have issued the **secure-mode enhanced** command to set the switch to enhanced secure mode.

Parameter	Description
`URL`	Specifies the URL of the container application. URL supports HTTP protocol. The image - location URL can either be IPv4 or IPv6 address. The IPv6 address must be provided within square brackets.
`vrf <VRF-NAME>`	(Optional) Specifies the VRF of the image URL.
`allow-unsigned`	(Optional) Allow download and deployment of an unsigned container image.

Examples

Configures the image location for the IPv4 setting:

```
switch(config-container-1)# image-location http://10.0.0.1/container.img
vrf mgmt
```

Appends the port to the address if the image server is running on a port other than HTTP for an IPv4 setting:

```
switch(config-container-1)# image-location http://10.0.0.1:9050/
container.img vrf mgmt
```

Configures image location for IPv6 setting by wrapping IP address between square brackets:

```
switch(config-container-1)# image-location http://[2001::2]/container.img
vrf mgmt
```

Specifies port number by appending it with the IPv6 address:

```
switch(config-container-1)# image-location http://[2001::2]:9050/
container.img vrf mgmt
```

When you include the **allow-unsigned** parameter on a switch in standard secure mode, the following message will be displayed to inform this can be a potential security issue.

```
switch(config-container-1)# image-location http://10.0.0.1/container.img
vrf mgmt allow-unsigned

Allowing unsigned container images poses a potential security risk
that can impact both the current device and the entire network. By
allowing installation of unsigned applications you are acknowledging
and accepting these risks. HPE shall not be responsible for the
```

```
consequences of your actions and disclaims any and all liability.
Continue (y/n)? y
```

When you attempt to include the **allow-unsigned** parameter on a switch in enhanced secure mode, the following message will appear to indicate that this parameter is not supported.

```
switch(config-container-1)# image-location http://10.0.0.1/container.img
vrf mgmt allow-unsigned
            Unsigned images are not permitted in the current secure mode,
using the
allow-unsigned parameter will have no effect.
```

Release	Modification
10.14	The **allow - unsigned** parameter is introduced.
10.12	Command introduced.

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8325H 8325P 8360 8400 9300 9300S 10000 10040	`config-container-<CONTAINER-NAME>`	Administrators or local user group members with execution rights for this command.

`ip address`

Syntax

```
ip address <ipaddr>/<submask>
no ip address <ipaddr>/<submask>
```

Description

Configures an IPv4 address and subnet mask for a virtual interface for an isolated network. Each virtual interface must have a unique, non-overlapping subnet. After completing the configuration, use the `show-container-isolated-network` command to verify that the subnet assignments are valid. This will automatically check for overlapping or duplicate subnets and report any detected conflicts.

The **no** version of this command removes the IP address configuration from the virtual interface.

> **NOTE**
>
> Do not assign the same IP address to both a virtual interface and an SVI (interface VLAN) in any VRF that uses the same VLAN. Doing so causes both the SVI and the virtual interface will respond to ARP requests for that IP address —potentially using different MAC addresses (the system MAC from the SVI or the container MAC from the virtual interface). This can lead to non-deterministic behavior in the network, as devices may send traffic to either the SVI or the container depending on which ARP reply they have cached. To ensure consistent connectivity and routing, always assign each IP address uniquely to either the SVI or the virtual interface, but not both.

Parameter	Description
`<ipaddr>`	An IPv4 address for the virtual interface. The IP address must be in CIDR notation (or example 192.0.2.0/24).

Example

Configuring the virtual interface with an IP address **10.0.0.2** and a subnet mask of **/24**.

```
switch(config-container-isolated-network-virtual-interface)# ip address
10.0.0.2/24
```

Command History

Release	Modification
10.17.1000	Additional description information.

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
8325 8325H 8325P	`config-container-isolated-network-<network-name>`	Administrators or local user group members with execution righ ts for this command.

`mdns enable`

Syntax

```
mdns enable
no mdns enable
```

Description

Allow a container in isolated network mode to use the multicast DNS (mDNS) protocol for the isolated network. When mDNS is enabled, containers can look up names on a local network without a regular DNS server. This allows devices on the same VLAN or subnet find container services and match hostnames to IP addresses using multicast.

The **no** version of this command disables this feature.

Usage

This configuration is not required if the mDNS based service discovery (mDNS-SD) feature is already configured globally and in the same VLANs used by the container.

Example

Enabling the mDNS

```
switch(config-container-isolated-network-net1)# mdns enable
```

Command History

Release	Modification
10.17	Command introduced

Command Information

Platforms	Command context	Authority
8325 8325H 8325P	`config-container-isolated-network-<network-name>`	Administrators or local user group members with execution righ ts for this command.

`mount`

Syntax

```
mount <id>
   destination <path>
   no ...
   source <TYPE> <PATH>
```

Description

Mount persistent storage for a container application, and optionally indicate the mounted storage source and destination paths.

Parameter	Description
<ID>	ID assigned to a mounted storage instance for a container appli cation. Range: <1 - 4294967295>
`destination` *<PATH>*	The destination path is a directory or file inside the container fil e system. The path must use the syntax **/{FILE|DIRECTORY}**. O nly letters, numbers, slashes, underscores, dashes, and periods are allowed.

Parameter	Description
`no ...`	Negates any configured parameter or removes a source or desti nation path.
`source`	Define the mounted storage source.
`<type>`	This version of AOS - CX supports only a **usb** storage type. **NOTE** For containers using USB persistent storage fun ctionality, **ext4** is the recommended formatting method for USB storage devices. While **FAT32** is currently supported, it may be deprecated in fut ure releases.
`<PATH>`	The mounted storage source path must be a valid file or directo ry on the USB device and is relative to the root directory of th e device file system. The path must use the syntax **/{FILE|DIRE CTORY}**. Only letters, numbers, slashes, underscores, dashes, a nd periods are allowed.

Example

Creating the mount ID 1 for the container **test**.

```
switch(config-container-test)# mount 1
```

Configuring container mount **1** with a **USB** source and the path **/app1-data**.

```
switch(config-container-mount-1)# source usb /app1-data
```

Configuring container mount **1** with the destination path **/app1-data**.

```
switch(config-container-mount-1)# destination /app1-data
```

Command History

Release	Modification
10.16.1000	Command introduced

Command Information

Platforms	Command context	Authority
8100	`config-container-`	Administrators or local user group members with execution righ ts for this command.

Platforms	Command context	Authority
8320 8325 8325H 8325P 8360 9300 9300S 10000 10040	*<CONTAINER-NAME>*	

`network isolated-network`

Syntax

```
network isolated-network <name>
no network isolated-network <name>
```

Description

Configure the isolated network to be used by the container application operating in isolated-network mode. The **no** version of this command ensures that the container no longer uses the indicated isolated-network as its network mode.

Parameter	Description
`<name>`	Name of the new isolated network

Example

Associate the container **app** with an isolated network named **net1**.

```
switch (config-container-app)# network isolated-network net1
```

Command History

Release	Modification
10.17	Command introduced

Command Information

Platforms	Command context	Authority
8325 8325H 8325P	`config-container-isolated-network-<network-name>`	Administrators or local user group members with execution righ ts for this command.

`network vrf`

Syntax

```
network vrf <VRF>
no network vrf <VRF>
```

Description

Use this command to associate a VRF with the container and allow connectivity through the VRF. When configured, the VRF will be accessible to the container through a network namespace.

Usage

Only one network VRF can be configured per container. If an additional VRF is configured, the container deployment will fail and generate an error message indicating that the container has more than one network VRF. Modifying a network container VRF will modify the container network configuration and the container will be restarted to apply the new changes if the container has already been enabled.

Examples

The following example defines the VRF **mgmt** for the **app** container.

```
switch(config-container-app)# network vrf mgmt
```

Command History

Release	Modification
10.16	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8325H 8325P 8360 8400 9300 9300S 10000 10040	`config-container-<CONTAINER-NAME>`	Administrators or local user group members with execution rights for this command.

`port-map`

Syntax

```
port-map host-port <PORT> container-port <PORT> protocol (tcp | udp)
```

Description

Issue this command from the **config-container-network-vrf-<name>** context to define a port map from a port on the host switch to a port on the container. This will allow communication from external devices to services provided by the container.

Parameter	Description
`host-port <port>`	The host switch port number. Range: 1 - 65535
`container-port <port>`	The container port number. Range: 1 - 65535
`tcp`	Communications will use the **TCP** protocol.
`udp`	Communications will use the **UDP** protocol.

Usage

A port map is an optional configuration for a container network VRF, but a port mapping is required to expose the mapped port to the outside network. This command can be issued multiple times to define more than one port map for the container.

> **NOTE**
>
> Host port conflicts are not allowed. The same host port cannot be used for different containers in the same VRF, and the host port used for the VRF cannot be reserved for a non-container application on the host switch.

Examples

Configure a port map from the host to the container for TCP traffic, where traffic to TCP port 8000 on the switch will be directed to TCP port 9000 in the container.

```
switch(config-container-network-vrf-mgmt)# port-map host-port 8000
container-port 9000 protocol tcp
```

The following example maps different ports on the host to the same container port for TCP traffic:

```
switch(config-container-network-vrf-mgmt)# port-map host-port 8000
container-port 9000 protocol tcp
switch(config-container-network-vrf-mgmt)# port-map host-port 8001
container-port 9000 protocol tcp
```

The following example maps the same port on the host switch to the same container port for TCP and UDP traffic:

```
switch(config-container-network-vrf-mgmt)# port-map host-port 8000
container-port 9000 protocol tcp
```

```
switch(config-container-network-vrf-mgmt)# port-map host-port 8000
container-port 9000 protocol udp
```

The following example maps the same port on different VRFs on the host switch to the same container port for TCP traffic:

```
switch(config-container-network-vrf-mgmt)# port-map host-port 8000
container-port 9000 protocol tcp
switch(config-container-network-vrf-default)# port-map host-port 8000
container-port 9000 protocol tcp
```

When removing a port mapping, if the command parameters do not match a configured port mapping, the command output displays the following message:

```
switch(config-container-network-vrf-mgmt)# no port-map host-port 8001
container-port 9000 protocol tcp
The specified port mapping is not configured
```

Command History

Release	Modification
10.16.1000	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8325H 8325P 8360 8400 9300 9300S 10000	`config-container-<CONTAINER-NAME>`	Administrators or local user group members with execution rights for this command.

Platforms	Command context	Authority
10040		

`restrict cpu`

Syntax

```
restrict cpu <PERCENTAGE>
no restrict cpu
```

Description

Configures limitations for the container CPU usage. The CPU constraint is set as a percentage of the total switch CPUs. One container or any combination of containers configured on a switch can use up to 20% of the total CPU capacity of the device.

> **NOTE**
> Configuring the CPU constraint for an already operational container will cause the container to restart if the container has already been enabled.

The **no** form of this command removes restrictions on the CPU usage, and reverts back to the default limitation of 10%.

Parameter	Description
<PERCENTAGE>	Specifies percentage for the container CPU usage, The default value is 10%.

Command History

Release	Modification
10.12	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8325H 8325P 8360 8400 9300 9300S 10000 10040	`config-container-<CONTAINER-NAME>`	Administrators or local user group members with execution rights for this command.

`restrict memory`

Syntax

```
restrict memory <MB>
no restrict memory
```

Description

Configures limitations for memory usage of the container. The memory constraint is set in MB, and the maximum 20% of the capacity of the device can be configured for any combination of containers running on the switch. Configuring the memory constraint for an already operational container restarts the container if the container has already been enabled.

The **no** form of this command removes restrictions on the memory usage and reverts back to the default limitation of 256 MB.

Parameter	Description
<MB>	Specifies the maximum memory usage in MB.The default value is 256 MB.

Command History

Release	Modification
10.12	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8325H 8325P 8360 8400 9300 9300S 10000 10040	`config-container-`*<CONTAINER-NAME>*	Administrators or local user group members with execution rights for this command.

`show container`

Syntax

```
show container [<CONTAINER-NAME>]|[isolated-network <name>]
```

Description

Shows the configuration and status information of the containers or container isolated networks running in the system. If the container or isolated network name is not specified, the output of this displays information for all containers or isolated networks. When a container name or isolated network name is specified, the output displays information specific to that container or isolated network.

Parameter	Description
`<CONTAINER-NAME>`	Specifies the name of the container for which information needs to be displayed
`isolated-network <name>`	(For 8325, 8325H or 8325P Switch series only) Specifies the n ame of the isolated network for which information needs to be displayed.

Usage

The **Container status** field in the output of this command can display any of the following status values:

- **initializing**: Container initialization and configuration verification is in progress.
- **waiting**: Container is waiting for configuration prerequisites to be met.
- **configuration failed**: User-specified container configuration failed.
- **network configuration failed**: Container network provisioning failed.
- **acquiring**: Container is acquiring an image.
- **paused**: Container has been paused.
- **acquire failed**: Container failed to acquire an image.
- **deploying**: Container is deploying.
- **operational**: Container is operational (deployed and running).
- **deploy** failed: Container failed to deploy.
- **run failed**: Container failed while running.

- **exited**: Container completed its execution and exited.
- **paused**: Container has been paused.

Examples

The following example shows information for a configured container with a network VRF with port mappings:

```
switch# show container
Container                  : app1
  Container status         : operational
  Manifest status          : success
  Image status             : verified
  Image version            : 1.0.0
  Image location VRF       : mgmt
  Image location URL       : http://30.0.0.2:8000/container.img
  CPU limit                : 10%
  Memory limit             : 512MB
  Environment variables:
    PYP=/usr/bin/python3
  Encrypted environment variables:
    encryptedVar1
    encryptedVar2
  Vrf NetworkS:
    VRF name  : default
    Port map  :
        8080:80/tcp
        8080:8080/udp
```

The following example shows additional error messages:

```
switch# show container
Container                    : app
  Container status           : configuration failed
  Config failure reason      : Multiple definitions of environment variable PYP
  Manifest status            : error
  Manifest status reason     : 'exec' file not found in container
  Image status               : verified
  Image version              : 1.0.0
  Image location VRF         : mgmt
  Image location URL         : http://30.0.0.2:8000/container.img
  CPU limit                  : 10%
  Memory limit               : 512 MB
  VRFs                       : mgmt
  Environment variables      :
    PYP=/usr/bin/python3
```

```
  Encrypted environment variables:
    PYP
    encryptedVar2
```

The following example shows a configured container without signature validation:

```
switch# show container app1
Container                : app1
  Container status       : operational
  Manifest status        : success
  Image status           : allowed without signature
  Image version          : 1.0.0
  Image location VRF     : mgmt
  Image location URL     : http://30.0.0.2:8000/container.img
  CPU limit              : 10%
  Memory limit           : 512 MB
  Environment variables:
    PYP=/usr/bin/python3
  Encrypted environment variables:
    encryptedVar1
    encryptedVar2
  VRF Networks:
    VRF name  : default
    Port map  :
      8080:80/tcp
      8080:8080/udp
```

Sample output with a configured container on a 8325 Switch series using a VRF network. The output of the **show container** command for an 8325/8325H/8325P switch series is slightly different than for other AOS-CX switch series in that it can include information about an isolated network configuration.

```
switch# show container
Container                : app
Container status       : operational
Manifest status        : success
Image status           : verified
Image version          : 1.0.0
Image location VRF     : mgmt
Image location URL     : http://192.0.2.2:8000/container.img
CPU limit              : 10%
Memory limit           : 512MB
Environment variables:
  PYP=/usr/bin/python3
Encrypted environment variables:
  encryptedVar1
  encryptedVar2
VRF networks:
```

```
  VRF name   : mgmt
  Port map   : n/a
Isolated network: net1
Mounts: n/a
```

The following example output for the **show container isolated-network** command shows details for isolated network **net1** with mDNS enabled, three virtual interfaces, and two containers, **test1** and **test2** using this isolated network. Note that only 8325, 8325H and 8325P switch series support containers in isolated mode..

```
switch# show container isolated-network net1
Isolated network         : net1
Status                   : network_provisioned
Status reason            : n/a
Default gateway          : n/a
mDNS                     : enabled
Virtual MAC              : 00:55:00:A1:B2:C3
Containers               :
   test1
   test2
Virtual interfaces       :
------------------------
VLAN  IP address
-----------------------
10    10.0.0.1/24
20    20.0.0.1/24
4094  192.168.134.252/24
```

The following example shows the output of the show containers command out when there are no configured containers:

```
switch# show container
No containers configured
```

Command History

Release	Modification
10.17	The output of this command can display information for isolated network config urations on 8325, 8325H and 8325P Switch series.
10.12	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8325H 8325P 8360 8400 9300 9300S 10000 10040	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show capacities containers`

Syntax

```
show capacities containers
```

Description

Shows the maximum number of containerized applications that can be configured in the system.

Examples

Shows maximum number of containerized applications that can be configured:

```
switch# show capacities containers
System Capacities: Filter CONTAINERS
Capacities
Name
```

```
 Value
---------------------------------------------------------------------------
-----------------
Maximum number of containerized applications configurable in the
system                         2
Maximum number of virtual interfaces in an isolated network configurable in
the system      64
```

Command History

Release	Modification
10.12	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8325H 8325P 8360 8400 9300 9300S 10000 10040	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show capacities-status containers`

Syntax

```
show capacities-status containers
```

Description

Reserved for future use.

Command History

Release	Modification
10.12	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8325H 8325P 8360 8400 9300 9300S 10000 10040	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show running-config container`

Syntax

```
show running-config container
```

Description

Shows the running configuration of all the containers.

Parameter	Description
`container`	Specifies that container running configuration must be displaye d.

Examples

Shows the running configuration for the container:

```
   container app1
       enable
    image-location http://30.0.0.2:8000/container.img vrf mgmt
    restrict cpu 10
    restrict memory 512
    network vrf mgmt
        port-map host-port 8000 container-port 9000 protocol tcp
        exit
    mount 1
        source usb /app1-data
        destination /app-data
        exit
    env PYP value /usr/bin/python3
    env encryptedVar1 encrypted ciphertext AQBapcmTCVdag
container app2
    enable
    image-location http://[2001::2]:8000/changeValidation_x86_t.img vrf
mgmt allow-unsigned
    restrict cpu 5
    restrict memory 256
    network vrf default
        port-map host-port 7819 container-port 7819 protocol udp
        exit
    mount 1
        source usb /app2-data1
        destination /app-data1
```

```
        exit
    mount 2
        source usb /app2-data2
        destination /app-data2
        exit
    env PYP value /usr/bin/python3
    env encryptedVar1 encrypted ciphertext AQBapcmTCVdag
    env encryptedVar2 encrypted ciphertext AQpY4V4v9UtDa==
```

Command History

Release	Modification
10.16.1000	The output of this command includes mount and network VRF details, if configured.
10.12	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8325H 8325P 8325P 8360 8400 9300 9300S 10000 10040	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`virtual-interface vlan`

Syntax

```
virtual-interface vlan <1-4094>
no virtual-interface vlan <1-4094>
```

Description

Sets up a virtual interface in an isolated network for a container in isolated-network mode. This virtual interface is linked to a VLAN ID and works as a network connection for the isolated network. It lets containers communicate with external devices connected to the switch's front panel ports within the same VLAN.

Parameter	Description
`<1-4094>`	A VLAN ID. Each isolated network virtual interface must be ass ociated with unique VLAN ID, and that VLAN must already exist on the switch.

Example

Configuring a virtual interface on vlan 10:

```
switch(config-container-isolated-network-<name>)# virtual-interface vlan 10
```

Command History

Release	Modification
10.17	Command introduced

Command Information

Platforms	Command context	Authority
8325 8325H 8325P	`config-container-isolated-network-<network-name>`	Administrators or local user group members with execution righ ts for this command.

`vrf`

Syntax

```
vrf <VRF-NAME>
```

Description

This command is deprecated. Use the network vrf command to define a VRF for a container.

Command History

Release	Modification
10.12	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8360 8400 9300 9300S 10000 10040		

Support and Other Resources

Access HPE Aruba Networking support and updates, and view warranty and regulatory information

Subtopics

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	**https://www.hpe.com/us/en/networking/hpe - aruba - networking - support - services.html**
AOS - CX Switch Software Documentation Portal	
HPE Aruba Networking Support Portal	**https://networkingsupport.hpe.com/home**
North America telephone	1 - 800 - 943 - 4526 (US & Canada Toll - Free Number) +1 - 650 - 750 - 0350 (Backup—Toll Number)
International telephone	**https://www.hpe.com/psnow/doc/a50011948enw**

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components

- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	**https://developer.arubanetworks.com/hpe - aruba - networking - aoscx/docs/about**
Airheads social forums and Knowledge Base	**https://community.arubanetworks.com/**
AOS - CX Software Technical Update channel on You Tube.	Videos on new features introduced in this release : **https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP - Q_UL3CskS**
HPE Aruba Networking Hardware Documentation and Translations Portal	
HPE Aruba Networking software	**https://networkingsupport.hpe.com/downloads https://networkingsupport.hpe.com/downloads**
Software licensing and Feature Packs	**https://licensemanagement.hpe.com/**
End - of - Life information	**https://networkingsupport.hpe.com/end - of - life**

Accessing Updates

You can access updates from the HPE Aruba Networking Support Portal at **https://networkingsupport.hpe.com**.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

https://networkingsupport.hpe./notifications/subscriptions (requires an active HPE Aruba Networking Support Portal account to manage subscriptions). Security notices are viewable without an HPE Aruba Networking Support Portal account.

Warranty Information

To view warranty information for your product, go to **https://www.arubanetworks.com/support-services/product-warranties/**.

Regulatory Information

To view the regulatory information for your product, view the Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products, available at **https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts**

Additional regulatory information

HPE Aruba Networking is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see **https://www.arubanetworks.com/company/about-us/environmental-citizenship/**.

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